Biodiversity Management

Land Sciences

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Biodiversity Management (A10022)

Short Title:	Biodiversity Management		
Faculty	Science and Computing		
Department	Land Sciences		
Cluster	Horticulture		
Author	NMCCARTHY		
Credits	5	**Level:**	Advanced

Description of Module / Aims

This module is a general introduction to the management and conservation of biological diversity. The module is divided into four parts. (1) This part provides an outline of the major elements of global biodiversity, its geographical distribution and evolution and examines the threats to biodiversity and the need to manage and conserve it. (2) This part focuses on conservation at species and population level, looking at the techniques and opportunities available for both in situ and ex situ conservation. (3) Introduces management and conservation issues at an ecosystem level, e.g. coastal, riparian, plantation forest, farmland etc. Protected areas form a critical conservation tool at this level, and the module examines the categories of protected areas, criteria for selection etc. The opportunities for habitat restoration and species reintroduction are also explored. (4) This last part reflects on challenges in the future and outlines possible solutions.

Programmes

MGTH-0015	BSc (Hons) in Land Management (in Agriculture) (WD_SBSAG_B)	1 / 2 / M
MGTH-0015	BSc (Hons) in Land Management (in Forestry) (WD_SBSFO_B)	1 / 2 / E
MGTH-0015	BSc (Hons) in Land Management (in Horticulture) (WD_SBSHO_B)	1 / 2 / E

Indicative Content

- An overview of biodiversity conservation - concepts and definitions
- Outline the threats to biodiversity and explain the need for conservation strategies
- In situ and ex situ methods for conservation of threatened species and small populations
- SACs, SPAs and pNHAs: their selection, protection and management
- Habitat restoration and species reintroduction

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Assess the concepts and definitions of biodiversity.
2. Interpret the patterns of national and international biodiversity, assess the threats to biodiversity and critique the need for conservation.
3. Compare and contrast the in situ and ex situ methods available for conservation.
4. Distinguish between the categories of protected areas, the criteria for their selection and management techniques.
5. Evaluate opportunities for conservation outside protected areas.
6. Distinguish the opportunities and methods for habitat restoration and species reintroduction.
7. Prepare a biodiversity management plan for a site.

Learning and Teaching Methods

- Lectures.
- Independent learning.

- Field trips.

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	
Project	70%	7
Case Studies	30%	1,2,3,4,5,6

Assessment Criteria

<40%: Very limited knowledge and understanding of the subject matter. Lacking in technical ability/skill and understanding to carry out practical/professional tasks. Failure to meet the objectives of projects. Unable to carry out or submit independent study and research.

40%-49%: Demonstrate a limited knowledge of the subject matter. Show some technical ability/skill in carrying out and reporting on practical/professional tasks. Attempt made to meet objectives and carry out independent study and research. Final result lacks balance and arguments undeveloped.

50%-59%: Demonstrate satisfactory general knowledge of the main issues within the subject. Logically and competently carry out practical/professional tasks. Satisfactory handling and awareness in carrying out independent study and research. Reporting and projects adequately structured and referenced.

60%-69%: Demonstrate sound knowledge of the subject matter. Show ability to analyse and logically argue in an effective and mature style. Demonstrate initiative and the ability to criticise methods used and appreciation of experimental design when carrying out practical/professional tasks. Projects and reporting coherent and soundly structured illustrating wide and in depth reading on subject coupled with the proper use of references.

70%-100%: Demonstrate authoritative handling of complex material and a highly developed and extensive understanding of the subject. Provide well focused analysis and convincing argument on the subject. Demonstrate imaginative approach, excellent technical ability, awareness, and aptitude in carrying out and reporting on practical/professional tasks. Show initiative, individual thought, and enthusiasm in self study and independent learning.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Tutorial	12	
Independent Learning	99	

Essential Material

- Gaston, K.J. and J.I. Spicer. _Biodiversity: An Introduction_. 2nd ed. UK: Blackwell Science, 2003.
- Glasson, J., R. Therivel and A. Chadwick. _Introduction to environmental impact assessment: principles and procedures, practice and prospect_. 4th ed. UK: Routledge, 2012.
- Groome, M.J., G.K. Meffe and C.R. Carroll. _Principles of conservation biology_. 3rd ed. US: Sinauer Associates Inc, 2012.

Requested Resources

- Lecture Room: Loose Seated